



Bishoy Labib

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Professional Summary

Multidisciplinary Mechatronics Engineer and MSc. student driven by a genuine interest in emerging technologies, complex problem-solving, and lifelong learning. My goal is to combine technical precision with a proactive, adaptable mindset to deliver sustainable, automated, and digitalized solutions in high-tech environments.

Education

MSc. in Interdisciplinary Computing | Specialization: Future Industries 10/2025 – Current
IT-U Austria
BSc. in Mechanical Engineering 09/2021 – 01/2025
Eötvös Loránd University, Hungary (GPA: 4.61/5.0)

- Faculty's "Outstanding Student Recognition" Award
- 3rd Place -Scientific Students' Associations Conference, "Optimization of medical device manufacturing."

Experience

ZALNER Engineering Kft., Hungary 09/2024 – 10/2025
Quality Engineering

- Enforced ISO 9001, ISO GPS, and ASME Y14.5 (GD&T) standards to ensure full compliance across automotive (IATF 16949), injection molding (ISO 20457), and electronics industries.
- Automated inspections by programming robotic arms, VMMs, CMMs, and 3D scanners; utilized GOM Inspect and PolyWorks for high-precision inspection, mesh processing, and reverse engineering.
- Optimized data-driven QA processes, achieving a 30% increase in VMM reliability by refining calibration protocols, 8D root cause analysis, and APQP/PPAP part acceptance.
- Oversaw the full equipment lifecycle, including installation, calibration, maintenance, while leading technical training for customers.

Specialized Field Assignments, Germany

- **Heilig & Schwab GmbH:** Completed specialized training in tool presetter retrofitting, calibration procedures, signal converters 10 mA, and 1 V_{PP}.
- **PMT GmbH:** Executed the manufacturing and assembly of customized CVMM machines, including granite structures, mechanical subsystems, axis assemblies, sensors, and optical equipment.

Mobilinter Kft., Manufacturing Engineering Intern 02/2024 – 08/2024

- Conducted Bachelor's Thesis on "Optimization of medical device manufacturing technology," ensuring compliance with EU MDR, ISO 13485, and DIN EN 10243 (hot die forging) using analytical, numerical, and experimental methods.
- Designed custom fixturing systems for 5-axis CNC machining of medical instruments (orthodontic pliers) using CAD/CAM/CAE modeling and digital process planning.

Vehicle Systems Maintenance (Independent), Hungary 07/2022 – 02/2024

- Performed complex electromechanical repairs and maintenance on high-pressure hydraulic and fuel systems, ensuring compliance with European road safety standards for vehicle inspections.
- Diagnosed sensor and mechanical failures (e.g., ABS, drivetrain) to maintain peak operational performance.

Languages

English: C1 German: A2

Competencies

Quality & Metrology:	GOM Inspect ZEISS Inspect PolyWorks CMM Programming GD&T 3D Scanning VMM Optical Systems ISO 9001 IATF 16949 8D APQP / PPAP V&V Lean Six Sigma
Simulation & Experiment:	FEA / CFD ANSYS Mechanical FMEA Abaqus MATLAB Digital Twins Sim-to-Reality Hardware-in-the-Loop (HIL) Simscape Unity VR/AR HCI
Robotics & Automation:	ROS2 PLC (Ladder, ST) TIA Portal URDF / XacRO / OpenUSD IIoT (MQTT) Robotic Arm Programming Sensor Fusion HMI Design SCADA Electro-Pneumatics Embedded Systems
Manufacturing & Fabrication:	CNC Machining (Multi-axis) Rapid Prototyping DfM/DfA/DfS Water Jet / Laser Cutting Process Planning Power & Hand Tools EdgeCAM Sinumerik Fusion CAM Tooling Polymers Fabrication Injection Molding
Data & Machine Learning:	Python (PyTorch, TensorFlow) C++ C MSSQL AI/ML Deployment Computer Vision GIT Linux/Bash Power BI Excel VBA DMIS G-code
Maintenance:	Predictive & Preventive Maintenance Root Cause Analysis (RCA) Vibration Analysis Thermal Imaging Hydraulics / Pneumatics Oscilloscope Wiring Soldering
Operations & Logistics:	Design Documentation Agile (Scrum) Kanban SAP ERP Sustainability Inventory Management Technical Training International Field Support Project Management

Certificates

CS50 Python	Harvard University
CS50 X: Fundamentals of Computer Science	Harvard University
MATLAB Onramp Course	MathWorks Academy
Volunteering at UNESCO Heritage Site	European Solidarity Corps
Nation BuildersLeadership Training	U.S. Embassy